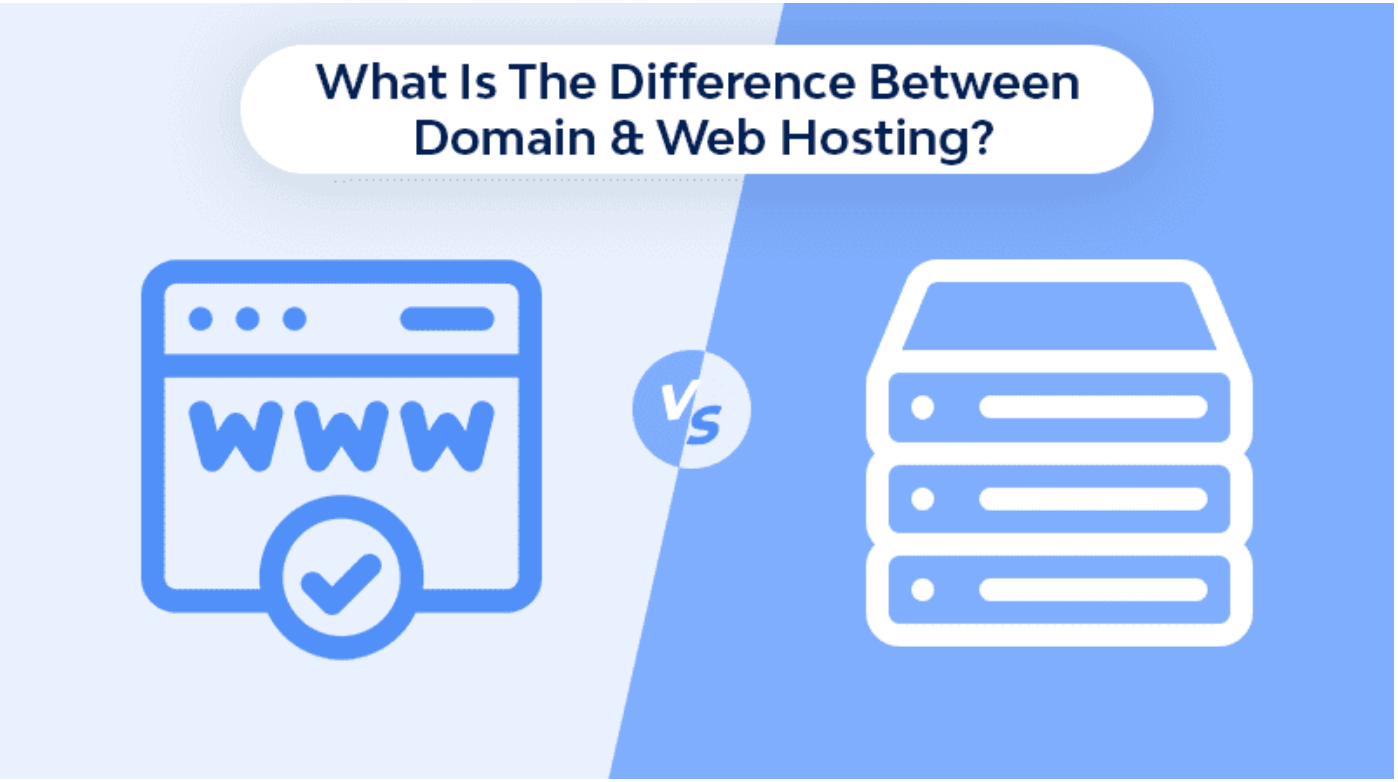
Domain Names, and Web Hosting



Introduction

Key Idea:
For a website to exist online, you need two main things:

- 🌐 **A Domain Name**
- 🏠 **Web Hosting**

These two work together to make a website accessible on the Internet.

What is a Domain Name?

Definition:
A **domain name** is the human-readable address of a website — easier to remember than an IP address.

Example:
- Domain name → cybercom.com
- IP Address → 10.45.0.192.201

Why it's useful:
- Easy to remember and share.
- Replaces complex numerical IP addresses.

Domain Name Structure

- **Example: labcyber.com**

Part	Description	Example
Second-Level Domain (SLD)	The main website name	labcyber
Top-Level Domain (TLD)	The extension	.com, .org, .net

Types of TLDs:

Type of TLD	Description	Examples
🌍 Generic TLDs (gTLDs)	Common domain extensions used globally	.com, .org, .net
GB Country Code TLDs (ccTLDs)	Represent specific countries or regions	.uk (United Kingdom), .us (United States), .in (India)
Sponsored TLDs (sTLDs)	Restricted domains for specific organizations or purposes	.edu (Education), .gov (Government), .mil (Military)

Domain Registration

Domain Registrars:

Companies that sell and manage domain names available for registration.

Example:

If your company name is **Byte Works**, you can buy byteworks.com from a domain registrar. (like GoDaddy or Namecheap), if it’s not already taken.

What is Web Hosting?

Definition:

A **web host** provides the **technologies and services** that allow a website to be accessible on the Internet.

They provide:

- Web servers
- Domain management
- Security & protection
- Backups
- Email accounts (e.g., alex@ byteworks.com)

Types of Web Hosting

There are 4 main hosting types:

Hosting Type	Description	Analogy
Shared Hosting	Many websites share one server. Cheapest but least powerful.	Like sharing a room with others.
Dedicated Hosting	One full server only for your website. Expensive but powerful.	Like owning a whole house.
VPS (Virtual Private Server)	Shared server but each site has its own private space.	Like having your own private room in a shared house.
Cloud Hosting	Uses many servers together to host websites. Scalable and reliable.	Like many houses working together to host guests.

Shared Hosting (Details)

- Most common and affordable
- Multiple sites share the same CPU, memory, and disk
- Performance can be affected by other websites

Use for: Personal blogs, small business sites.

Dedicated Hosting (Details)

- One full server only for you
- High performance and security
- Most expensive option

Use for: Large organizations or high-traffic e-commerce sites.

VPS Hosting (Details)

- Combines features of shared and dedicated hosting
- Each user has a private environment within one server
- Mid-range cost

Use for: Medium businesses needing flexibility and more control.

Cloud Hosting (Details)

- Websites are hosted on multiple connected servers
- Reliable — if one server fails, another takes over
- Scalable — easily handle more visitors

Use for: Modern scalable apps, growing startups.

Can You Host Your Own Website?

✅ Yes, it’s possible.
🚫 But not recommended — because:

- You must maintain the server 24/7.
- You handle all security and backups.
- It’s costly and time-consuming.

Better option: Use a professional web hosting provider.

Summary

- Websites need **a Domain Name + Web Hosting.**
- Domain = Website name (easier than IP address).
- Web hosting = Where website files live.
- 4 Hosting Types → Shared, Dedicated, VPS, Cloud.
- Use hosting companies for performance, security, and reliability.